

CodingLab

Final Presentation

SWE-4404

SPL-II ● Group-10

Team Members

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Project Overview : CodingLab

A web-based coding practice platform built for university students with role-based access,automated judging and full progress tracking

How it works

1



Solve

Pick a Problem

2



Code

Write Code in IDE

3



Submit

Run against Tests

4



Verdict

AC/WA/TLE/Error

Student Features

Discover Problems

Browse problems; filter by difficulty or search by title and topic tag

In-Browser Code Editor

Write and submit codes directly with syntax highlighting

Instant Judge Verdict

Jdoodle evaluation with verdicts, time & memory stats

Progess Dashboard

View total solved, success rate, difficulty stats, and progress charts

History

View submission history

Leaderboard

Compare rank against classmates - motivating healthy academic competition

Problem Setter Features



Create Problems

Define problem details, test cases, and execution limits.



Submit Problems for Review Approval

Submit problems for review, track status, and update from feedback.

Admin Features



Moderate Problems

Review pending problems and edit or remove inappropriate ones.



Manage Users

Manage users and role changes, and approve or reject Problem Setter requests.



Create & Manage Contests

Create, edit, and delete contests with selected problems and durations.

Project Structure

CodingLab

Client/Server split

frontend/ and backend/ deploy independently

Backend layers

routes → middleware → controllers → services → models

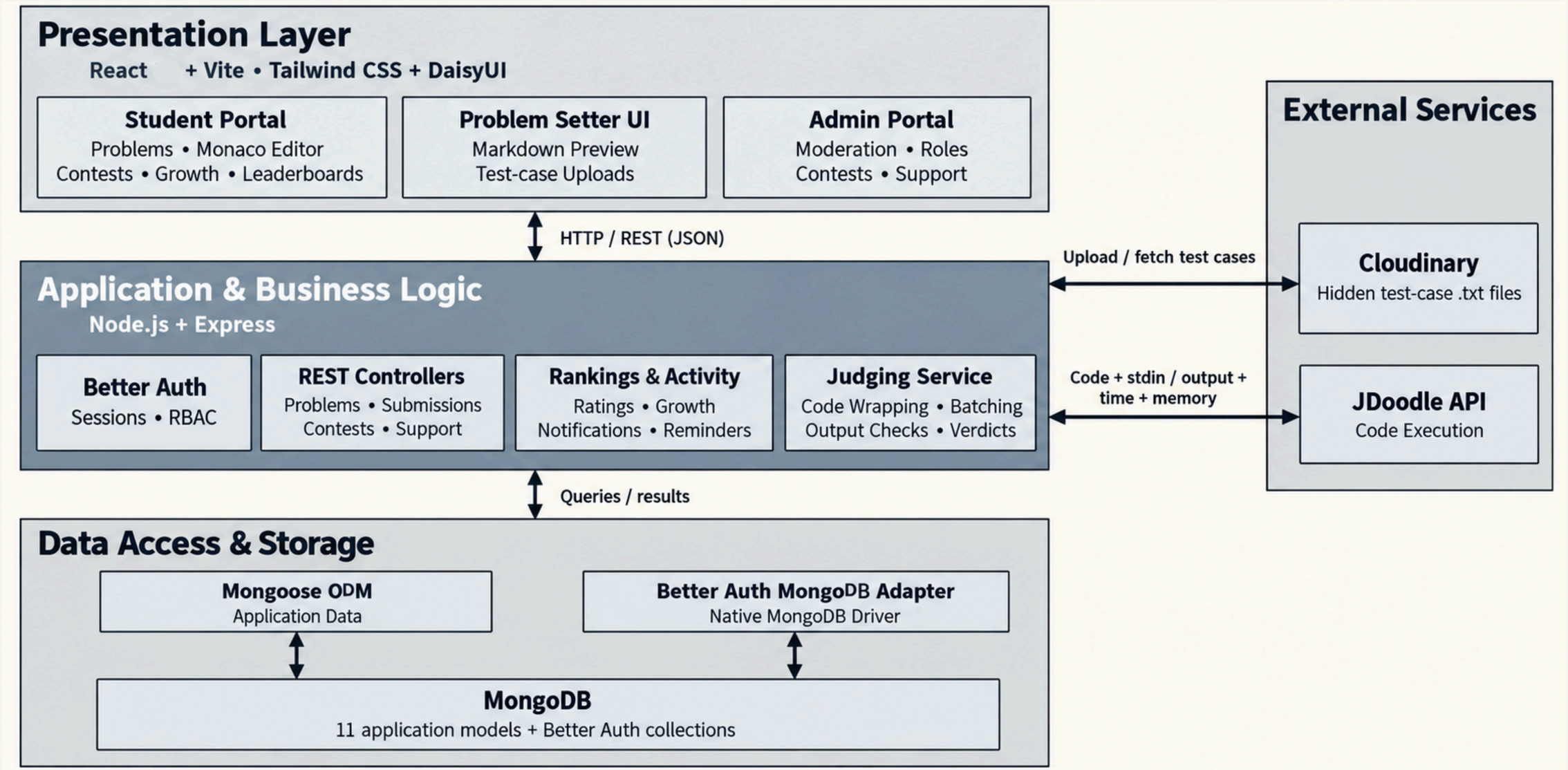
Frontend by role

(Student/Setter/Admin), components, hooks, lib

```
CodingLab/
└─src/
    ├─backend/
    │   ├─routes/          # 9 Express API routers (/api/*)
    │   ├─middleware/      # auth.js, accountAccess.js (RBAC)
    │   ├─controllers/     # request & response handlers
    │   ├─services/        # judgingService.js (JDoodle batch)
    │   ├─models/          # 11 Mongoose schemas (MongoDB)
    │   ├─lib/             # Better Auth, Cloudinary, JDoodle
    │   ├─utils/           # Standings, runtime wrappers
    │   └─tests/           # 27 backend test suites
    └─frontend/
        └─src/
            ├─pages/       # Student, Setter & admin/ views
            ├─components/  # workspace/ (Monaco), growth/ (charts)
            ├─hooks/       # useProblemEditor, useGrowthData
            └─lib/         # API fetch, run-code, auth-client
```

System Architecture

CodingLab



Classes and Methods

Functions

How the code is divided

- React function components for pages and UI
- 10 controllers handle API requests
- Judging service handles batching and verdicts
- Utilities handle Elo, standings and activity
- 11 Mongoose models define application data
- 1 custom class: JudgingConfigurationError

Why this structure

- Routes connect endpoints to controllers
- Middleware checks sessions and permissions
- Judging runs before submission persistence
- Shared utilities reuse ranking and growth logic
- Integration helpers configure external services

Size (lines of code)

Functions & methods: 911

Average 18.8 • Median 4 • Max 670

Custom classes: 1

JudgingConfigurationError: 7 lines

Source modules: 104 files

Average 132.7 • Max 678

Total source: 13,802 lines

Team Contribution

CodingLab

Raiyan Al Arafat
ID 230042134

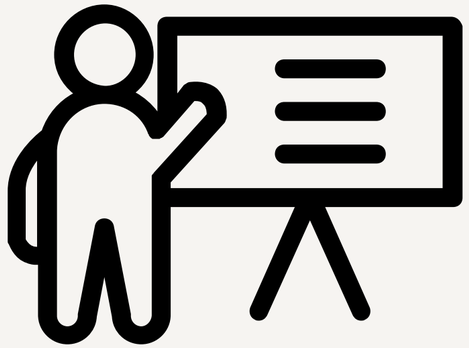
- UI & Design System
- Frontend Architecture
- Problem Submission Workflow
- Setter Onboarding

Rakibur Rahman Ananta
ID 230042128

- Elo-Rank Leaderboard
- User Growth Analytics
- Profile Dashboard

Nafian Nazat Mahid
ID 230042144

- Contest Architecture
- Admin Control Panel
- System Moderation Workflow



Demonstration...


Thank
You